Modelling Software-based Systems

Lecture 4

Validation, Verification and Proof Tools for Event-B for Correct-by-Construction

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General Summary

1. Proof Management
2. Development Methodology and Tools for Correct By Construction
3. Project Management

Current Summary

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Proof Management

- The panel called *proof control* is providing tools for *discharging* proof obligations
- Proof obligations are stated as follows : $\Gamma \vdash P$: *When assertions of* Γ *are supposed to hold, then* P *does hold*
- Problem : Checking that the sequent $\Gamma \vdash P$ is valid.
- How : Applying automatic procedures as pp, pr etc
- . . . or helping the proof tool to build the *proof tree*

Main Limits of PP (Predicate Prover)

- Quite sensitive to useless hypotheses.
- Unable to automatically abstract some statements.
- Sometimes poor management of equalities.
- Needs some creative hints

Creative Hints

- Proposing clever lemma : the user may have interesting ideas because he/she knows the problem to solve.
- Suggesting a proof by case.
- Proposing a witness for an existential goal.
- Proposing a universal hypothesis instanciation.
- etc.

Sequents and Inference rules

- Sequent (where $list_of_hypotheses$ might be empty).

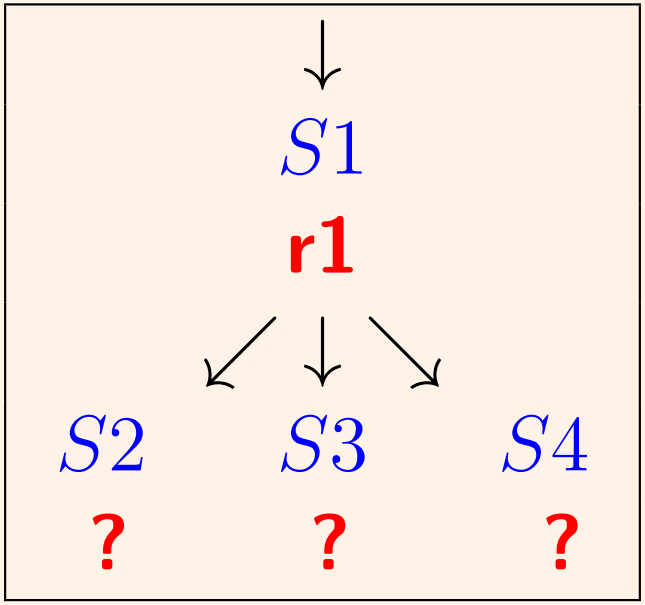
$$list_of_hypotheses \quad \vdash \quad conclusion$$

- Inference rule (where $list_of_sequents$ might be empty).

$$\frac{list_of_sequents}{sequent}$$

- Backwards application of inf. rules $\rightsquigarrow$ proof tree.

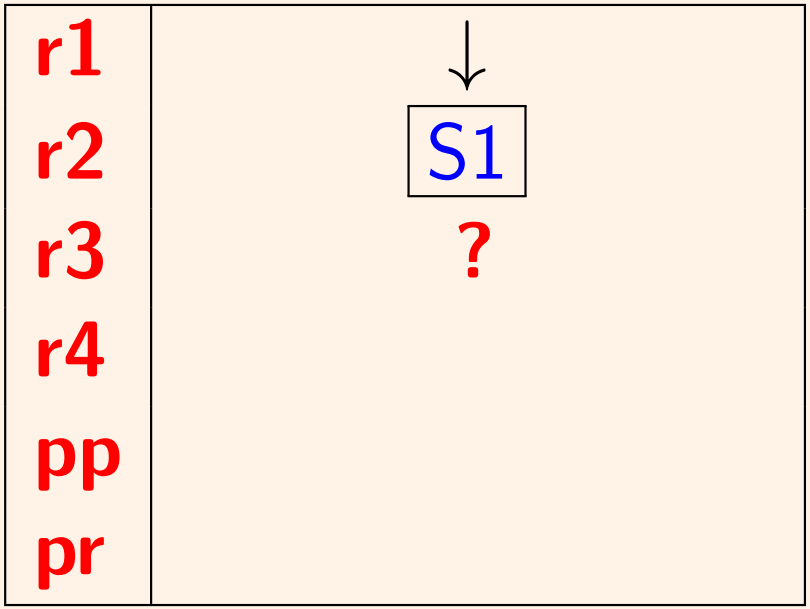
The Proof Tree



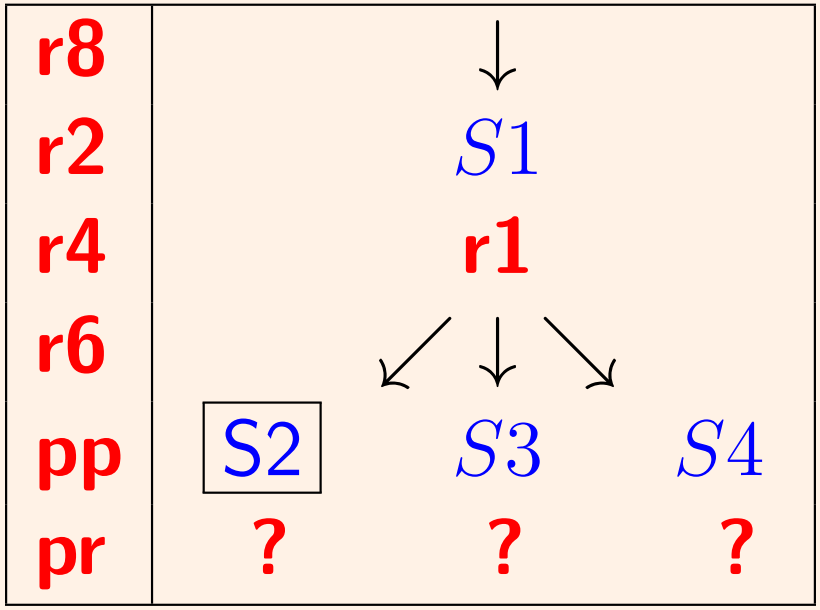
- Rule **r1** is applied to sequent S1.

- Producing sequents S2, S3, and S4.

Interface : Tree and Palette of Rule Buttons

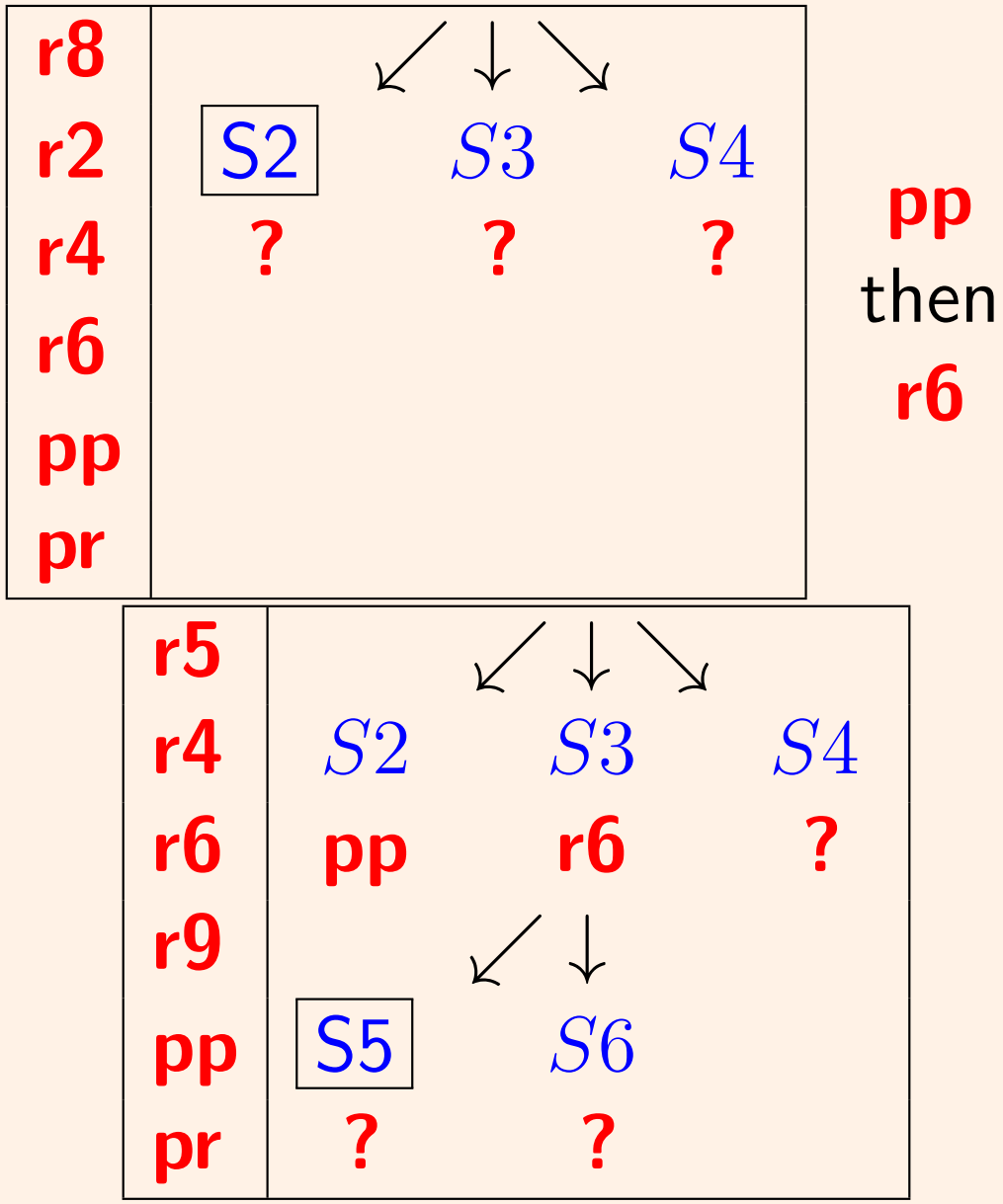


Pressing
r1



- The proof tree is constructed interactively.

Difficulty : Size of Window



- Our tentative "interface" does not work.

More Precise Form of Sequent

$$H \vdash L \Longrightarrow G$$

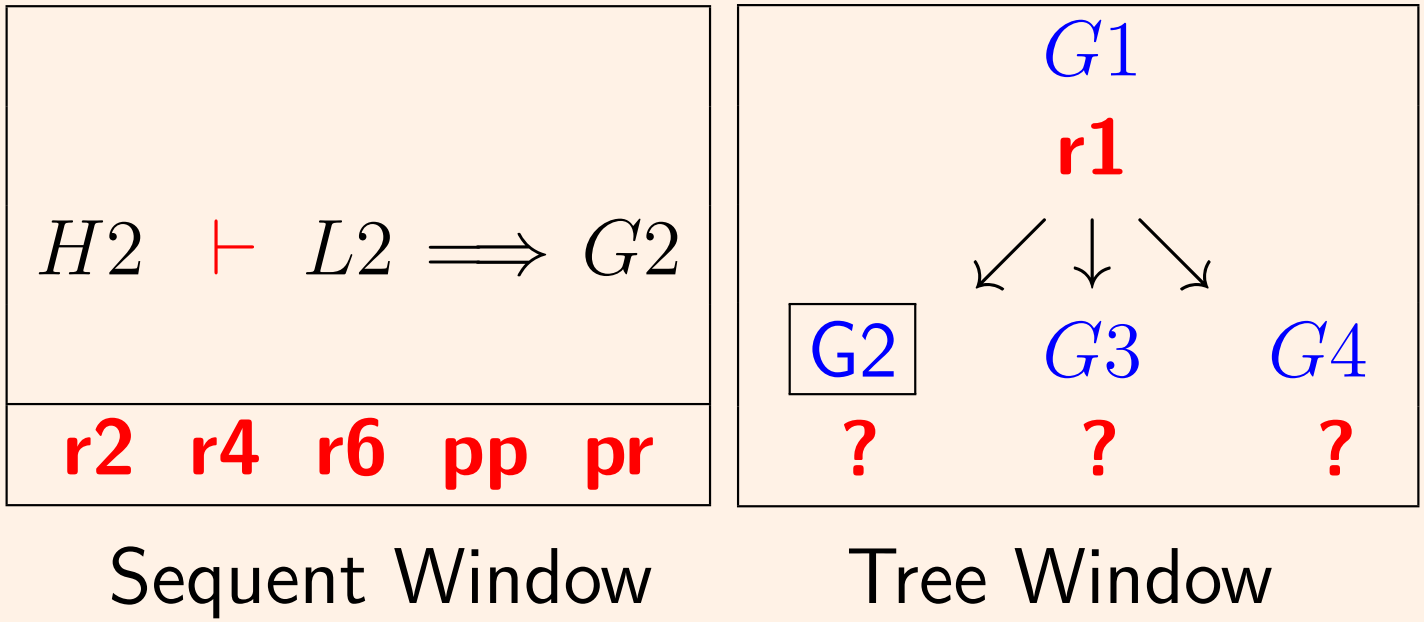
- Most of the time the conclusion has an implicative form
- G is called the goal (it is a non-conjunctive predicate)
- L are called the local hypotheses (might be empty)

$$H \vdash I(x) \wedge G(x) \wedge P(x, x') \Longrightarrow I(x')$$

Tentative Solution : Using Two Windows

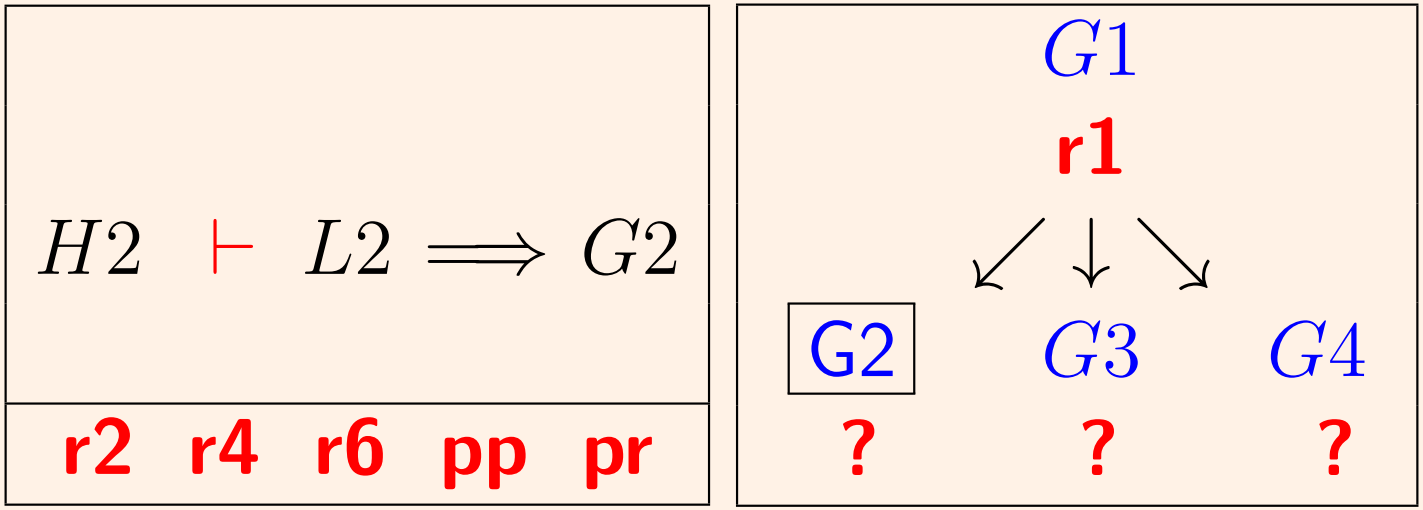
- Tree window with simplified sequents (goals only)

- Sequent window containing the full sequent of interest

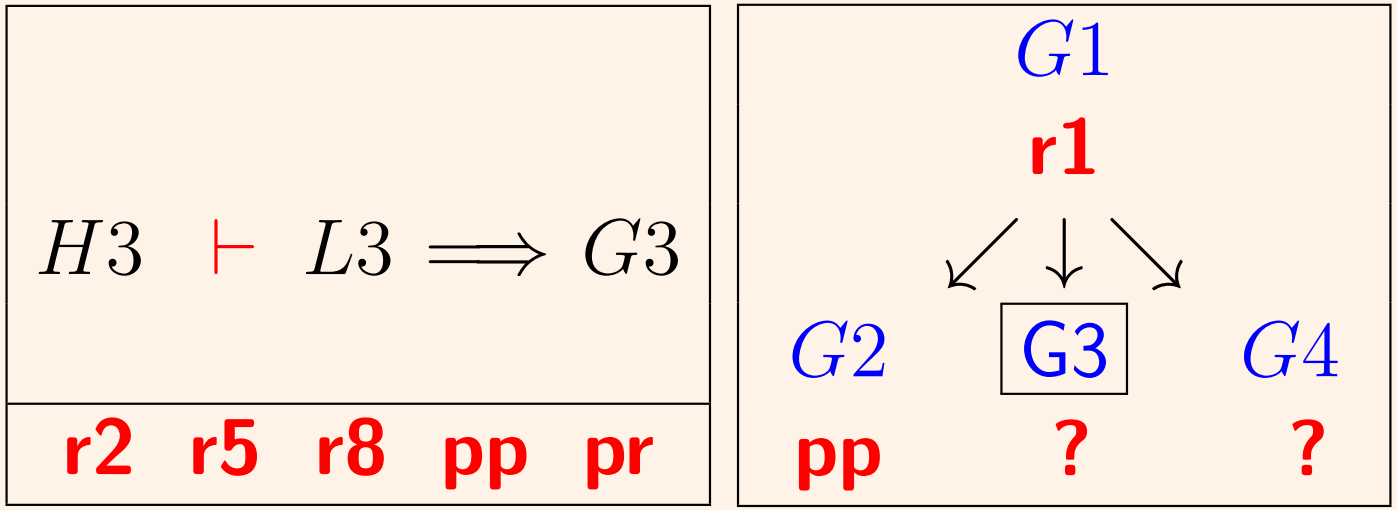


Sequent Window Tree Window

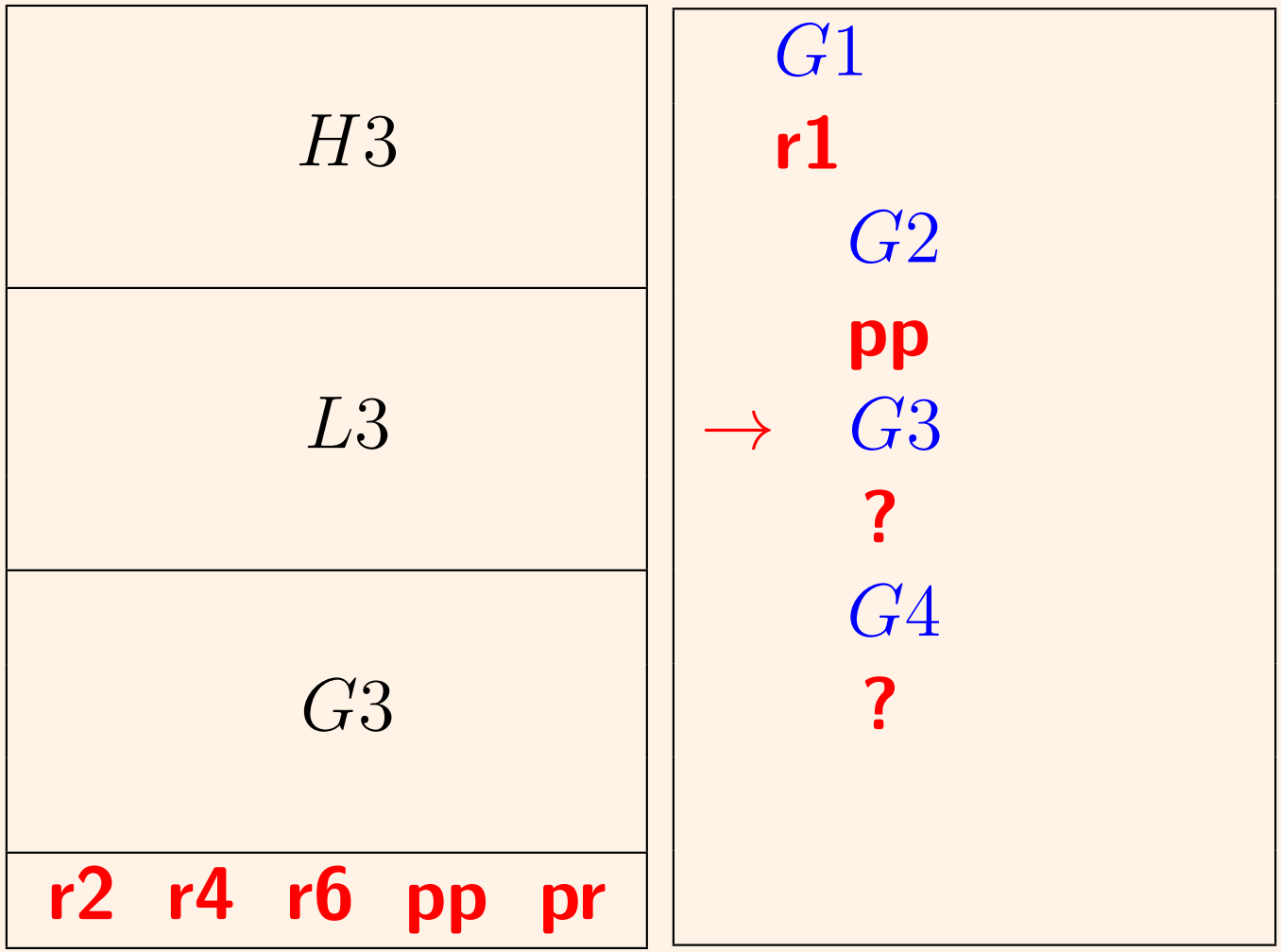
One Proof Step



Pressing **pp**



More Realistic Windows



More Elaborate Sequents

$$\underbrace{hidden \;;\; searched \;;\; cached}_{list_of_hypotheses} \vdash \underbrace{local \Longrightarrow goal}_{conclusion}$$

- $hidden$ Not visible on the sequent window
- $searched$ Visible after some search in $hidden$
- $cached$ Visible but not part of the conclusion any m
- $local$ Visible and part of the conclusion

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Context and Objectives

Context

Developing life-cycle methodology and a set of associated techniques and tools to develop the highly critical systems using formal techniques from requirements analysis to code generation.

Objectives

- To establish a unified theory for the critical system development.
- To Build a set of tools for supporting new development life-cycle methodology.
- To develop a closed-loop system for verification purpose.
- Graphical based refinement technique to handle the complexity of the system.
- To satisfy requirements and metrics for certifiable assurance and safety.
- To support evidence-based certification.

Main Contributions

Development Methodology

- Formal Methods based Critical System Development Life-Cycle

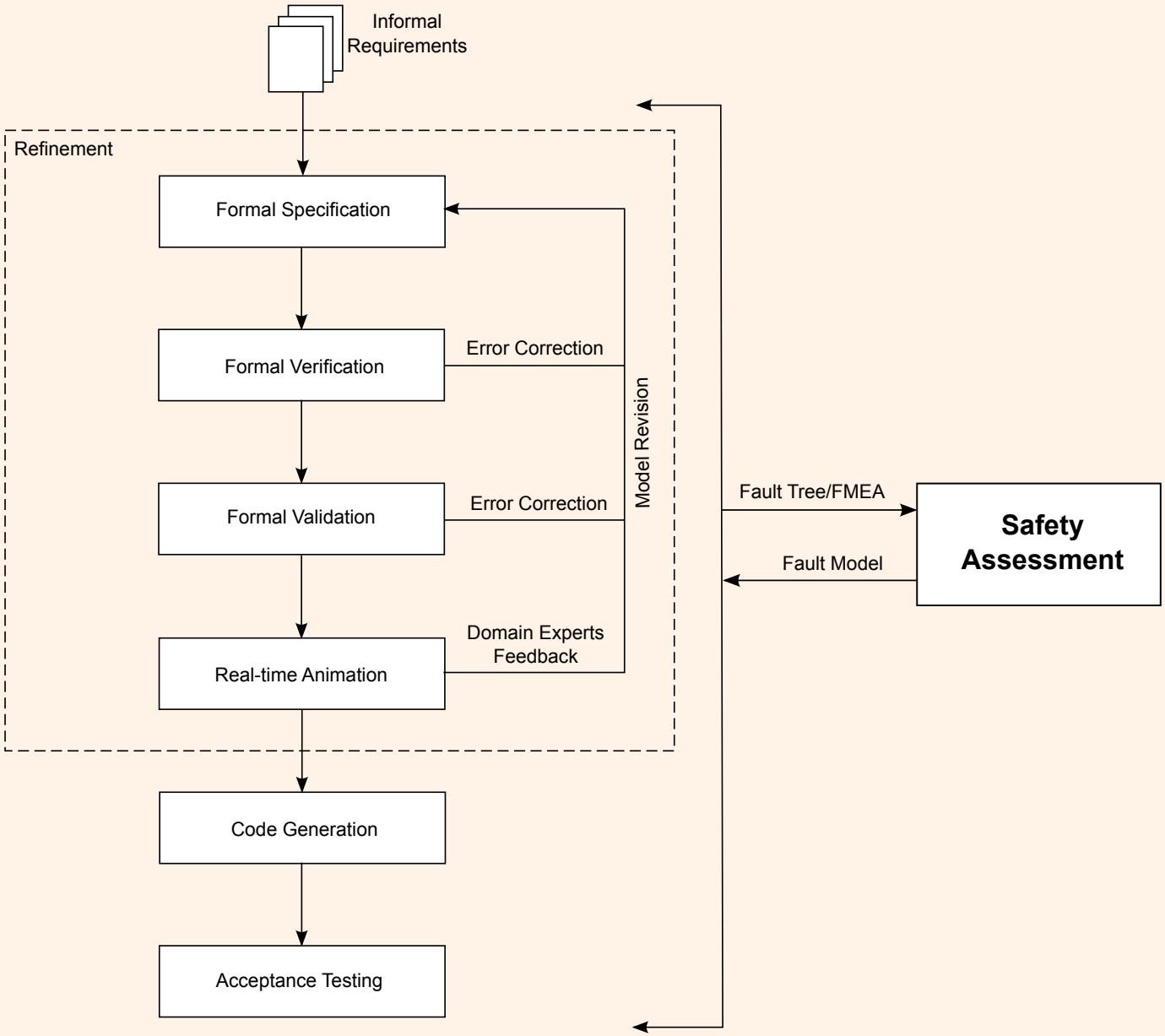
Techniques and Tools

- Real-Time Animator
- Refinement Chart
- Automatic Code Generator : EB2ALL
- Formal Logic Based Heart-Model

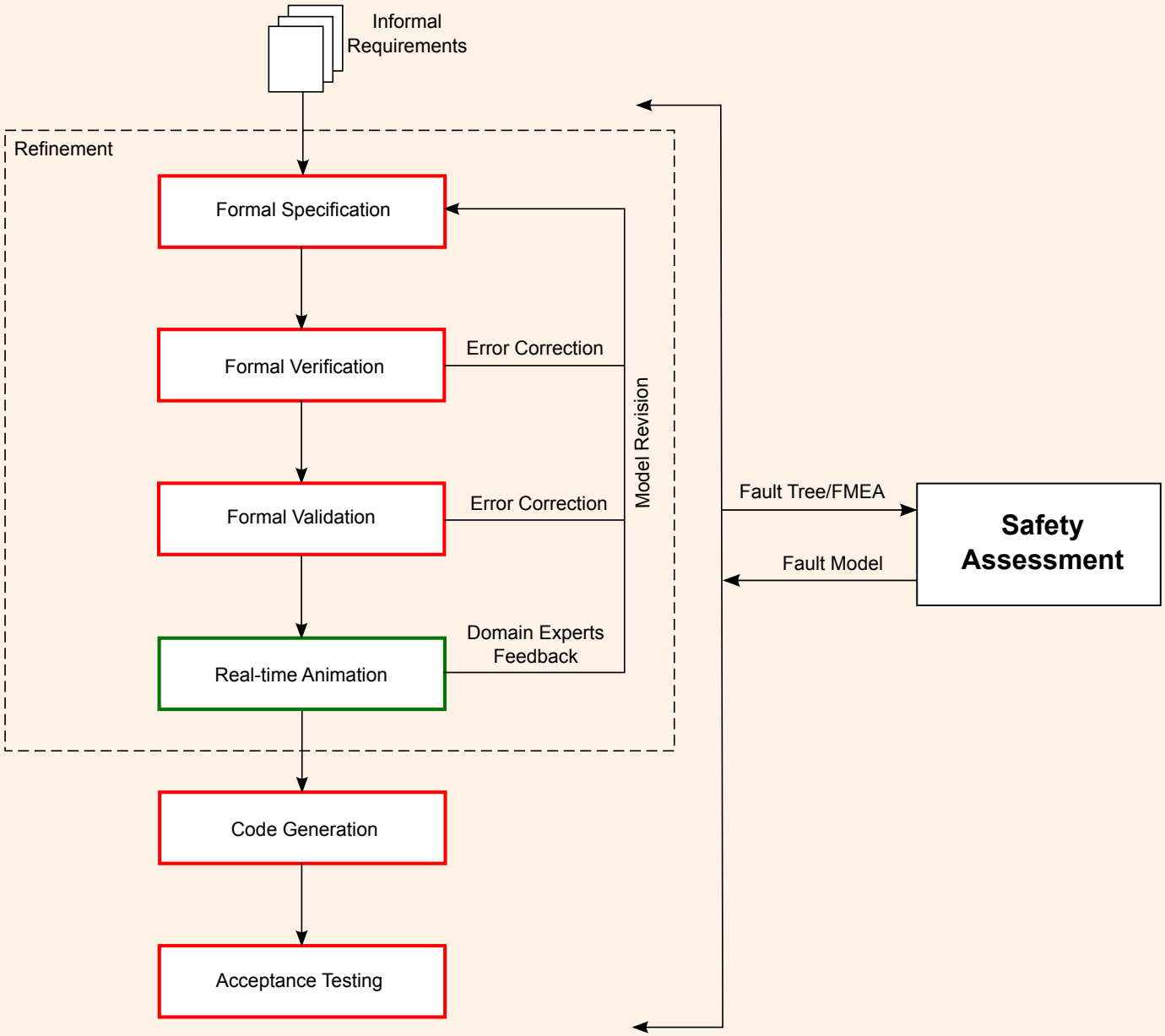
Case Studies

- The Cardiac Pacemaker

Critical System Development Life-Cycle Methodology



Critical System Development Life-Cycle Methodology



Real-Time Animator

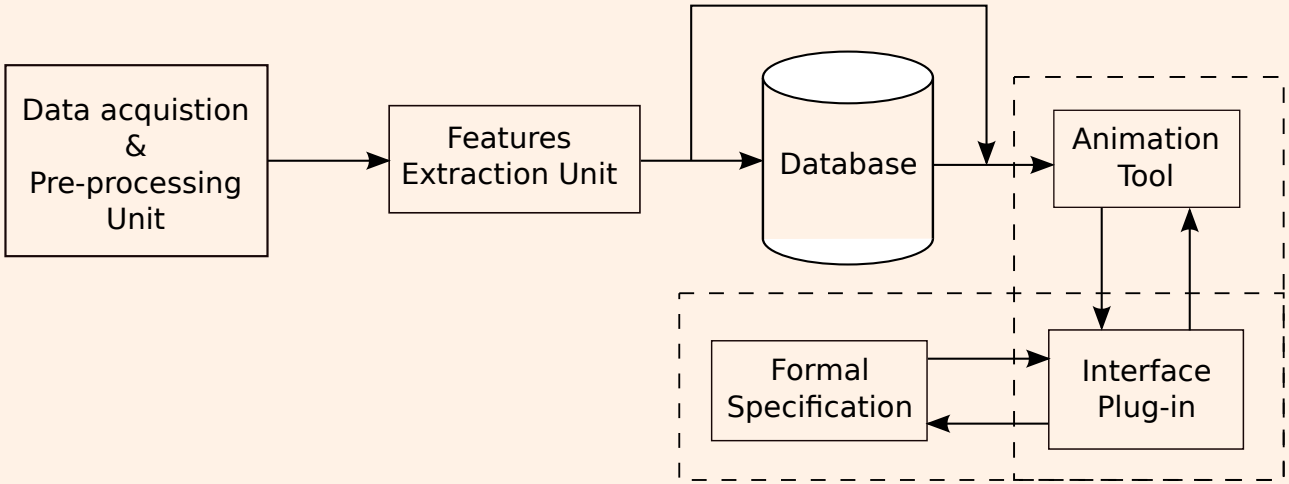
What is Real-time Animator ?

- Visual representation of formal model using real time data set.

Why should we use Formal Model Animator ?

- To validate system behavior according to the stakeholders
- To express formal models for non-mathematical domain experts
- To discover the error in the early stage of system development (Traceability)

Proposed Architecture



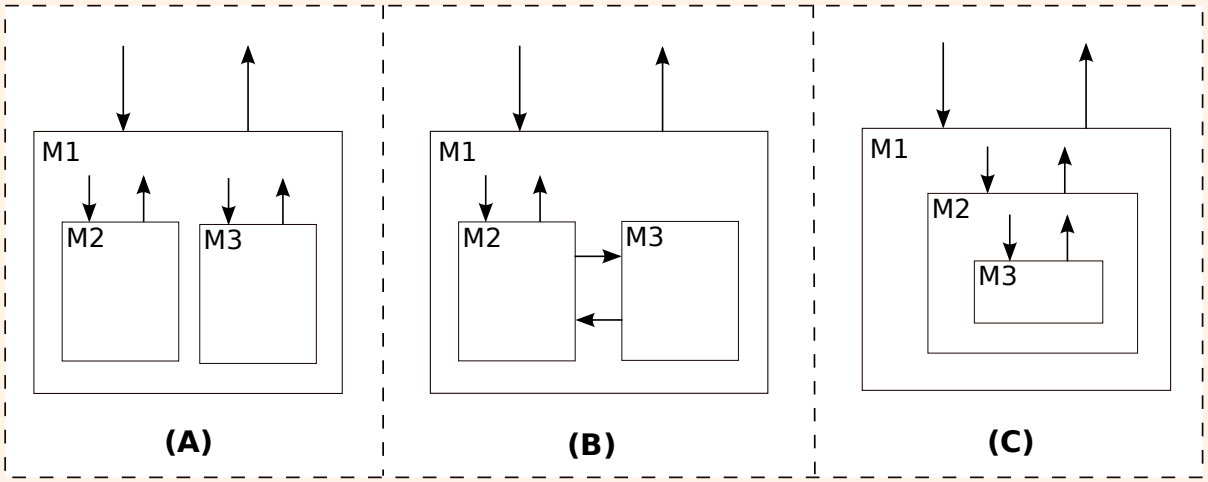
Refinement Chart

- Refinement chart is a graphical representation of a complex system using layering approach.
- Integration and features based code structuring of different components of the systems.

Basic rules of refinement chart

- Parallel Refinement
- Sequential Refinement
- Nested Refinement

$$M1 \sqsubseteq (M2 \parallel M3 \parallel ..... \parallel M_{n-1} \parallel M_n)$$
$$M1 \sqsubseteq (M2 \succ M3 \succ ..... \succ M_{n-1} \succ M_n)$$
$$(M1 \sqsubseteq M2,\ M2 \sqsubseteq M3, ... ,\ M_{n-1} \sqsubseteq M_n)$$



$$M1 \sqsubseteq (M2 \parallel M3)$$
$$M1 \sqsubseteq (M2 \succ M3)$$
$$(M1 \sqsubseteq M2,\ M2 \sqsubseteq M3)$$

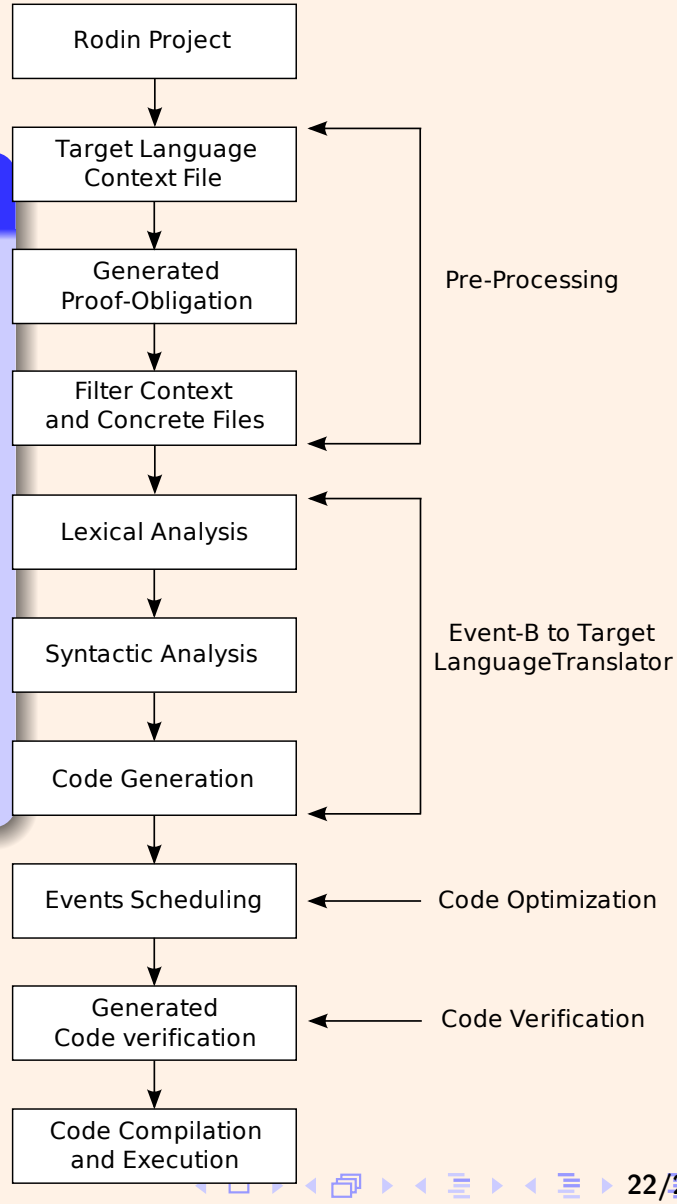
EB2ALL Code Generation Tool

EB2ALL

- Automatic code generation from Event-B formal specification to multiple languages (C, C++, Java, C#) to make it executable.
- A module for Rodin (Deploy is an European Commission FP7 project)
- It is more than 20000 lines of code
- More than 1000 downloads

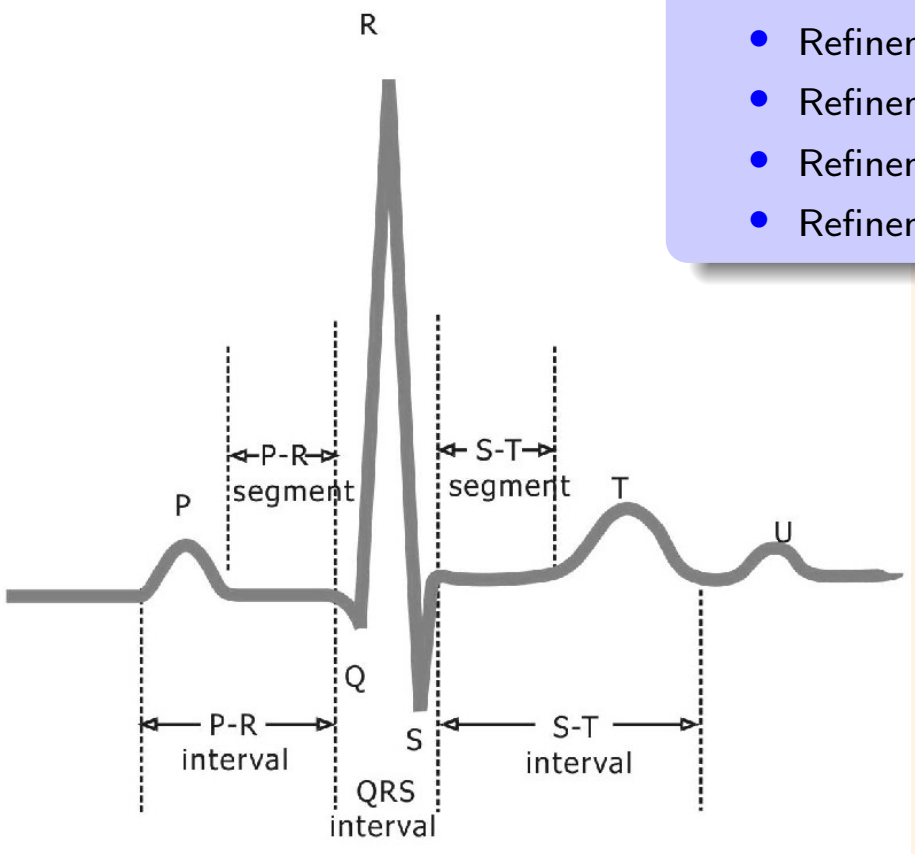
Download : http://eb2all.loria.fr/

EB2ALL Tool Architecture



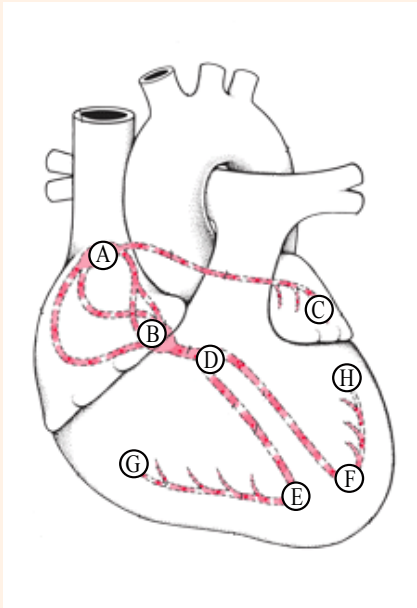
Formal Logic Based Heart-Model

Electrocardiogram (ECG)



Formal Development

- Abstract Model
- Refinement 1 : Introducing Steps in the Propagation
- Refinement 2 : Impulse Propagation
- Refinement 3 : Perturbation the Conduction
- Refinement 4 : Getting a Cellular Model



Landmarks on the Electrical Conduction of Heart

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Steps of the project

- Reading the description of the problem to solve.
- List the safety properties as partial correctness, mutual exclusion, ... and properties on sets, constants, ...
- Identify the state variables for the first machine
- Define inductive definitions if necessary and state correctness properties as $\forall n.n\mathbb{N} \Rightarrow n * n = v(n)$.
- List Use Cases : for instance, adding a new user, updating a a phone number, recording a movement ...
- First expression of a context and a machine.

Documentation

- **Property**
  - *the set of people is non empty and finite*
  - $people \neq \varnothing \wedge finute(people)$